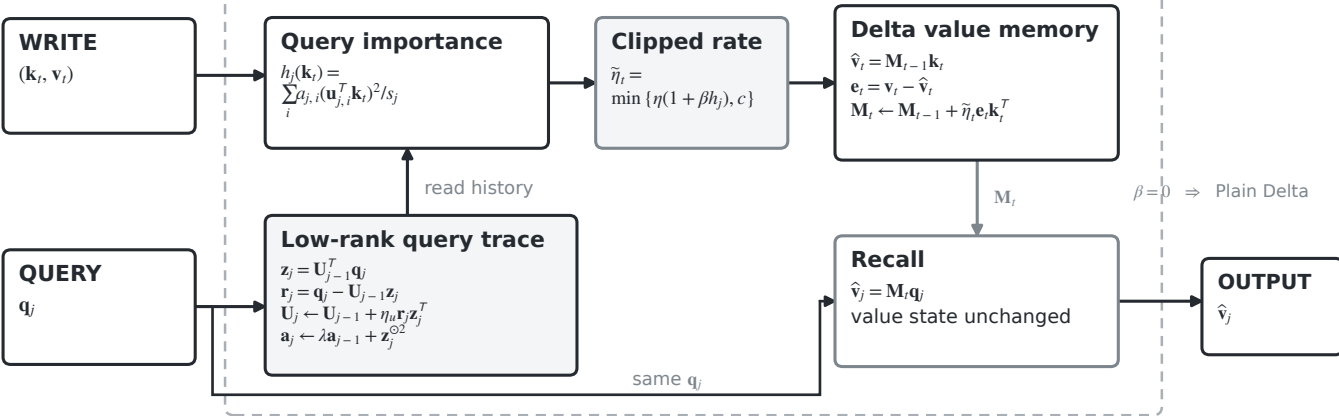
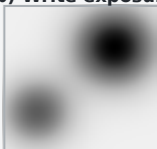


(a) Query Dual Dynamics Delta (QD³)



(b) Write exposure



non-uniform writes

×

Query-driven rate



$\tilde{\eta}(\mathbf{k})$ from queries

→

Effective allocation



$p_w(\mathbf{k}) \tilde{\eta}(\mathbf{k})$